

Course 4

Generated subspace, linear maps

↳ special kind of morphism

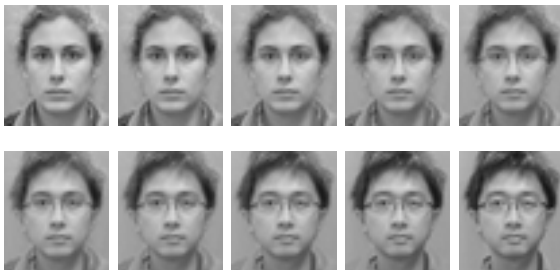


Prof. dr. Septimiu Crivei

- 1 Basic properties ✓
- 2 Subspaces ✓
- 3 Generated subspace
- 4 Linear maps

Following [Klein], we describe a way to achieve an image crossfade effect.

Linear algebra



Subspace $\left\{ \begin{array}{l} S \neq \emptyset \\ \dots \end{array} \right. \rightarrow$ rewritten in a single one $S \subseteq V$

For a vector space V over K , we denote by $S(V)$ the set of all subspaces of V . Sometimes, this set is denoted by $S_K(V)$ if we like to emphasize the field K .

Let V be a vector space over K and let $(S_i)_{i \in I}$ be a family of subspaces of V . Then $\bigcap_{i \in I} S_i \in S(V)$. I the set of $\mathbb{N}$

Proof. For each $i \in I$, we have $S_i \in S(V)$, hence $0 \in S_i$. Then $0 \in \bigcap_{i \in I} S_i \neq \emptyset$. Now let $k_1, k_2 \in K$ and $x, y \in \bigcap_{i \in I} S_i$. Then $x, y \in S_i, \forall i \in I$. But $S_i \in S(V), \forall i \in I$. It follows that $k_1x + k_2y \in S_i, \forall i \in I$, hence $k_1x + k_2y \in \bigcap_{i \in I} S_i$. Therefore, $\bigcap_{i \in I} S_i \in S(V)$. $\rightarrow$ condition of the subspace.

The $\cap$ is still a subspace.

Is the 0 in there?

$\cap$ $\left\{ \begin{array}{l} \text{dual concepts} \\ - \text{comm} \\ - \text{assoc} \end{array} \right.$ $\square$

U of subspaces is not generally a subspace.

In general, the union of two subspaces of a vector space is not a subspace. For instance, the sets

$\mapsto 0x$

$$S = \{(x, 0) \mid x \in \mathbb{R}\},$$

$\mapsto 0y$

$$T = \{(0, y) \mid y \in \mathbb{R}\}$$

// The same as ...

are subspaces of the canonical real vector space $\mathbb{R}^2$, but $S \cup T$ is not a subspace of $\mathbb{R}^2$. Indeed, for instance, we have

$(1, 0), (0, 1) \in S \cup T$, but $(1, 0) + (0, 1) = (1, 1) \notin S \cup T$.

The origin is a subspace; any line containing the origin is a subspace.

12 vector space; take subspaces in there $\rightarrow$ not subspace.

Now we are interested in how to "complete" a given subset of a vector space to a subspace in a minimal way.

Can you form the smallest...?

Let V be a vector space and let $X \subseteq V$. Then we denote

$$\langle X \rangle = \bigcap \{S \leq V \mid X \subseteq S\}$$

$\uparrow$ minimal set
 the intersection of all subspaces containing that subset.

$\hookrightarrow$ it is a subspace

and we call it the subspace generated by X or the subspace spanned by X . Here X is called the generating set of $\langle X \rangle$.

finite n. of vectors

If $X = \{v_1, \dots, v_n\}$, we denote $\langle v_1, \dots, v_n \rangle = \langle \{v_1, \dots, v_n\} \rangle$.

(1) $\langle X \rangle$ is the "smallest" (with respect to inclusion) subspace of V containing X . (each of the members contain X)

(2) $\langle \emptyset \rangle = \{0\}$. (subspace generated by the empty set)

$\rightarrow$ complete this fig. in a similar way to become convex.



(3) If $S \subseteq V$, then $\langle S \rangle$ is the smallest subspace of V containing S .

Who is the smallest... containing S ?

A vector space V over K is called **finitely generated** if

$\exists v_1, \dots, v_n \in V$ ($n \in \mathbb{N}$) such that

(finite nr. of vectors)

$$V = \langle v_1, \dots, v_n \rangle. \quad \text{precisely the whole space.}$$

Then the set $\{v_1, \dots, v_n\}$ is called a **system of generators** for V .

Let V be a vector space over K and $v_1, \dots, v_n \in V$ ($n \in \mathbb{N}$). A finite sum of the form

$$\underline{k_1 v_1 + \dots + k_n v_n}, \quad \leftarrow \text{a formal expression like this}$$

where $k_i \in K$ ($i = 1, \dots, n$), is called a (finite) **linear combination** of the vectors $v_1, \dots, v_n$. $\leadsto$ FOR THE SUBSPACE $k_1 v_1 + k_2 v_2$

↳ more practical than the orig. stuff

Let V be a vector space over K and let $\emptyset \neq X \subseteq V$. Then

Subspace generated by X : we cannot compute all subspaces

$$\langle X \rangle = \{k_1 v_1 + \dots + k_n v_n \mid k_i \in K, v_i \in X, i = 1, \dots, n, n \in \mathbb{N}^*\},$$

that is, the set of all finite linear combinations of vectors of X .

Proof. We prove the result in 3 steps, by showing that

$$L = \{ \underline{k_1 v_1 + \dots + k_n v_n} \mid k_i \in K, v_i \in X, i = 1, \dots, n, n \in \mathbb{N}^* \}$$

scalar
↓
(0 · N)

$\langle X \rangle$ is the smallest subspace of V containing X .

v_1, v_2 non-collinear

is 0 there? Yes. $\checkmark$ (0 · N)
adding them $\Rightarrow$ linear combination $\checkmark$
is $\forall$ a linear combination? Yes. $\checkmark$

$\Rightarrow$ the whole plane is the smallest subspace.

(i) Let $v \in X$. Then $v = 1 \cdot v \in L$, hence $L \neq \emptyset$. Now let $k, k' \in K$ and $v, v' \in L$. Then $v = \sum_{i=1}^n k_i v_i$ and $v' = \sum_{j=1}^m k'_j v'_j$ for some $k_1, \dots, k_n, k'_1, \dots, k'_m \in K$ and $v_1, \dots, v_n, v'_1, \dots, v'_m \in X$. Then $kv + k'v' \in L$, because it is a finite linear combination of vectors of X . Hence we have $L \leq V$. *L the smallest subset containing X.*

(ii) If $v \in X$, then $v = 1 \cdot v \in L$. Hence $X \subseteq L$.

(iii) Let $S \leq V$ be such that $X \subseteq S$. Let $k_1, \dots, k_n \in K$ and $v_1, \dots, v_n \in X$. Since $X \subseteq S$ and $S \leq V$, $\underbrace{k_1 v_1 + \dots + k_n v_n}_{\text{elements of } L} \in S$. Hence $L \subseteq S$.

Thus, we have $\langle X \rangle = L$ by the initial remark.

Induction

□

Let V be a vector space over K and let $x_1, \dots, x_n \in V$. Then

Apply the Theorem above: *all the finite linear combinations.* *finite no. of vectors*
 $\langle x_1, \dots, x_n \rangle = \{k_1 x_1 + \dots + k_n x_n \mid k_i \in K, x_i \in X, i = 1, \dots, n\}.$

Stk. to prove in the exam, this is not the easiest.

(a) Consider the canonical real vector space $\mathbb{R}^3$. Then

unitary vectors $\rightarrow$ what do they generate?

$\langle (1, 0, 0), (0, 1, 0), (0, 0, 1) \rangle$ triples. How do we add? Component-wise.

$$= \{k_1(1, 0, 0) + k_2(0, 1, 0) + k_3(0, 0, 1) \mid k_1, k_2, k_3 \in \mathbb{R}\}$$

$$= \{(k_1, 0, 0) + (0, k_2, 0) + (0, 0, k_3) \mid k_1, k_2, k_3 \in \mathbb{R}\}$$

$$= \{(k_1, k_2, k_3) \mid k_1, k_2, k_3 \in \mathbb{R}\} = \mathbb{R}^3. \text{ only some special vectors will generate that.}$$

Hence $\mathbb{R}^3$ is generated by the three vectors $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$, and thus it is **finitely generated**.

(b) Consider the canonical vector space $\mathbb{Z}_2^3$ over $\mathbb{Z}_2$. Similarly as above, we have:

$$\langle (\hat{1}, \hat{0}, \hat{0}), (\hat{0}, \hat{1}, \hat{0}) \rangle = \{(k_1, k_2, \hat{0}) \mid k_1, k_2 \in \mathbb{Z}_2\} \neq \mathbb{Z}_2^3.$$

$\hookrightarrow$ they generate sth. smaller

not the whole space

Hence $\mathbb{Z}_2$ is not generated by the two vectors $(1, 0, 0)$ and $(0, 1, 0)$. But it is generated by $(\hat{1}, \hat{0}, \hat{0})$, $(\hat{0}, \hat{1}, \hat{0})$ and $(\hat{0}, \hat{0}, \hat{1})$, hence it is finitely generated.

(c) Consider the subspace

→ contains 0 $\Rightarrow$ is a subspace.
→ equation of a plane

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid x - y - z = 0\}$$

of the canonical real vector space $\mathbb{R}^3$. Let us write it as a generated subspace. Expressing $x = y + z$, we have:

$$\begin{aligned} S &= \{(\overbrace{y+z}^{\text{arrow}}, y, z) \mid y, z \in \mathbb{R}\} = \{(y, y, 0) + (z, 0, z) \mid y, z \in \mathbb{R}\} \\ &= \{\cancel{y}(1, 1, 0) + \cancel{z}(1, 0, 1) \mid y, z \in \mathbb{R}\} = \langle (1, 1, 0), (1, 0, 1) \rangle. \end{aligned}$$

Consider the linear comb. of those vectors $\hookrightarrow$ the set of all linear comb.

Alternatively, one may express y or z by using the other two components and get other writings of S as a generated subspace, namely $S = \langle (1, 1, 0), (0, -1, 1) \rangle = \langle (1, 0, 1), (0, 1, -1) \rangle$. We see

! A subspace may be generated by $\neq$ vectors.

[Very big project $\rightarrow$ break it down: vector spaces could be complicated
 $\rightarrow$ break it apart]

In what follows we shall be interested in “decomposing” a vector space into subspaces.

Definition 1.1. Let V be a vector space over K and let $S, T \leq V$.

We define the **sum** of the subspaces S and T as the set

$$S + T = \{s + t \mid s \in S, t \in T\}.$$

each vector from s + each vector from t .

$\rightarrow$ subspace generated by the **UNION**

Lemma 1.2. Let V be a vector space over K and $S, T \leq V$. Then

$$S + T = \langle S \cup T \rangle, \text{ hence } S + T \leq V. \rightarrow \text{subspace.}$$

Proof. We prove the equality by double inclusion.

First, let $\underline{v = s + t} \in S + T$, for some $s \in S$ and $t \in T$. Then

vector written in
this form

$$v = 1 \cdot s + 1 \cdot t \quad \text{linear comb. of vectors from the } U.$$

is a linear combination of the vectors $s, t \in S \cup T$, hence $v \in \langle S \cup T \rangle$. Thus, $S + T \subseteq \langle S \cup T \rangle$.

Now let $v \in \langle S \cup T \rangle$. Then separate.

$$\underline{v} = \sum_{i=1}^n k_i v_i = \underbrace{\sum_{i \in I} k_i v_i}_{\text{all vectors from } S} + \underbrace{\sum_{j \in J} k_j v_j}_{\text{all vectors from } T} = \underline{\text{linear comb. of vectors from } U}$$

linear comb. of vectors from the U .

where $I = \{i \in \{1, \dots, n\} \mid v_i \in S\}$ and

$J = \{j \in \{1, \dots, n\} \mid v_j \in T \setminus S\}$. But the first sum is a linear combination of vectors of S , hence it belongs to S , while the second sum is a linear combination of vectors of T , hence it belongs to T . Thus, $v \in S + T$ and so $\langle S \cup T \rangle \subseteq S + T$. $\square$

The $\cap$ of subspaces must contain at least 0.

Let V be a vector space over K and let $S, T \leq V$.

If $S \cap T = \{0\}$, then $S + T$ is denoted by $S \oplus T$ and is called the *direct sum* of the subspaces S and T .
 $\hookrightarrow$ extra property: $S \cap T = \{0\}$

Let V be a vector space over K and let $S, T \leq V$. Then

$$V = S \oplus T \iff \overset{\text{unique writing}}{\forall v \in V, \exists! s \in S, t \in T : v = s + t.}$$

Proof. $\Rightarrow$ Assume that $V = S \oplus T$. Let $v \in V$. Then $\exists s \in S, t \in T$ such that $v = s + t$. Now suppose that $\exists s' \in S, t' \in T$ such that $v = s' + t'$. Then $s + t = s' + t'$, whence
 $\hookrightarrow v$ written in 2 $\neq$ ways

$$\underset{ES}{s} - \underset{ES}{s'} = \underset{ET}{t'} - \underset{ET}{t} \in S \cap T = \{0\}.$$

Hence $s = s'$ and $t = t'$, that show the uniqueness.

$\Leftarrow$ Assume that $\forall v \in V, \exists! s \in S, t \in T$ such that $v = s + t$.
Then $V \subseteq S + T$. Clearly, we have $S + T \subseteq V$ and consequently
 $V = S + T$. Now suppose that $0 \neq v \in S \cap T$. Then
take a non-zero vector in the $\cap$.

$$v = \underbrace{v}_{\in S} + \underbrace{0}_{\in T} = \underbrace{0}_{\in S} + \underbrace{v}_{\in T} \Rightarrow 2 \text{ different writings of } v \Rightarrow \text{not true.}$$

But this is a contradiction, since we have the uniqueness of writing of v as a sum of an element of S and an element of T . Therefore,
 $S \cap T = \{0\}$ and thus, $V = S \oplus T$. $\square$

Consider the canonical real vector space $\mathbb{R}^2$. Then $\mathbb{R}^2 = S \oplus T$,
where $S = \{(x, 0) \mid x \in \mathbb{R}\}$ and $T = \{(0, y) \mid y \in \mathbb{R}\}$. *direct sum*

$\hookrightarrow O_x$

$\hookrightarrow O_y$

Can be decomposed on 2 axes

↳ „Aplicatii liniare“

Let V and V' be vector spaces over the same field K . A function $f : V \rightarrow V'$ is called:

(1) $(K-)$ linear map (or (vector space) homomorphism or linear transformation) if

$\Rightarrow$ $\neq$ group homomorphism of...

$\rightarrow$ $\neq$ compatible with the addition of vectors

DEF: $\left\{ \begin{array}{l} f(v_1 + v_2) = f(v_1) + f(v_2), \quad \forall v_1, v_2 \in V, \\ f(kv) = kf(v), \quad \forall k \in K, \forall v \in V. \end{array} \right.$

$\rightarrow$ compatible with the multiplication of scalar w vector?

$\nearrow$ Rings

linear map

(2) *isomorphism* if it is a bijjective K -linear map, same domain, codom

(3) *endomorphism* if it is a K -linear map and $V = V'$.

(4) *automorphism* if it is a bijjective K -linear map and $V = V'$.

↳ See A.

special function between vector spaces.

If $f : V \rightarrow V'$ is a K -linear map, then the first condition from its definition tells us that f is a group homomorphism between the groups $(V, +)$ and $(V', +)$. Then we have $f(0) = 0'$ and $f(-v) = -f(v)$, $\forall v \in V$. underlying abelian group

We denote by $V \overset{\text{isomorphic}}{\simeq} V'$ the fact that two vector spaces V and V' are isomorphic. We also denote

$$\text{Hom}_K(V, V') = \{f : V \rightarrow V' \mid f \text{ is } K\text{-linear}\}, \text{ isomorphisms}$$

$$\text{End}_K(V) = \{f : V \rightarrow V \mid f \text{ is } K\text{-linear}\}, \text{ set of all endomorphisms}$$

$$\text{Aut}_K(V) = \{f : V \rightarrow V \mid f \text{ is bijective } K\text{-linear}\}. \text{ automorphisms}$$

→ In practice, in showing that sth. is a K -linear map.

Let V and V' be vector spaces over K and $f : V \rightarrow V'$. Then f is a K -linear map $\iff f(k_1v_1 + k_2v_2) = k_1f(v_1) + k_2f(v_2), \forall k_1, k_2 \in K, \forall v_1, v_2 \in V$. $\neq$ compatible with finite linear combinations.

Proof. $\implies$ Let $k_1, k_2 \in K$ and $v_1, v_2 \in V$. Then

$$f(k_1v_1 + k_2v_2) = f(k_1v_1) + f(k_2v_2) = k_1f(v_1) + k_2f(v_2).$$

$\impliedby$ Choose $k_1 = k_2 = 1$ and then $k_2 = 0$ to get the two conditions of a K -linear map. $\square$

→ using the definition

takes all the elements from the domain to 0.

(a) Let V and V' be vector spaces over K and let $f : V \rightarrow V'$ be defined by $f(v) = 0'$, $\forall v \in V$. Then f is a K -linear map, called the **trivial linear map**.

(b) Let V be a vector space over K . Then the **identity map** $1_V : V \rightarrow V$ is an **automorphism** of V . $\rightarrow$ subspace

(c) Let V be a vector space and $S \leq V$. Define $i : S \rightarrow V$ by $i(v) = v$, $\forall v \in S$. Then i is a K -linear map, called the **inclusion linear map**. $\leadsto$ identity map but with a $\neq$ domain - ?

(d) Let V be a vector space over K and $a \in K$. Define $t_a : V \rightarrow V$ by $t_a(v) = av$, $\forall v \in V$. Then **t_a is an endomorphism of V** .

$\hookrightarrow$ extension / compression

(e) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = x + y$. Then f is an $\mathbb{R}$ -linear map, because we have

$$\begin{aligned}
 f(\underbrace{k_1(x_1, y_1)}_{\text{pairs}} + \underbrace{k_2(x_2, y_2)}_{\text{pairs}}) &= f(k_1x_1 + k_2x_2, k_1y_1 + k_2y_2) \\
 &= (k_1x_1 + k_2x_2) + (k_1y_1 + k_2y_2) \\
 &= k_1(x_1 + y_1) + k_2(x_2 + y_2) \\
 &= k_1f(x_1, y_1) + k_2f(x_2, y_2)
 \end{aligned}$$

$\nearrow$ assume the canonical structure (component-wise)
EXAMPLES!
Check if it is linear map.

for every $k_1, k_2 \in K$ and for every $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$.

On the other hand, $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = xy$ is not an $\mathbb{R}$ -linear map, because, for instance, we have

$$f((1, 0) + (0, 1)) = f(1, 1) = 1 \neq 0 = f(1, 0) + f(0, 1).$$

Not a linear map.

(f) Let $\theta \in \mathbb{R}$ and let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

which is the counterclockwise rotation of angle θ about the origin in the plane. Then f is an $\mathbb{R}$ -linear map. In particular, for $\theta = \frac{\pi}{2}$, we have $f(x, y) = (-y, x)$. *effect of this linear map $\rightarrow$ rotating $\rightarrow$ comp. graphics*

(g) For an interval $I = [a, b] \subseteq \mathbb{R}$ we considered the real vector space

Analysis

$$\mathbb{R}' = \{f \mid f : I \rightarrow \mathbb{R}\} \text{ all functions}$$

and its subspaces

$$C(I, \mathbb{R}) = \{f \in \mathbb{R}' \mid f \text{ continuous on } I\},$$
$$D(I, \mathbb{R}) = \{f \in \mathbb{R}' \mid f \text{ derivable on } I\}.$$

Do we have linear maps for this? Yes.

Then

$$F : D(I, \mathbb{R}) \rightarrow \mathbb{R}', \quad F(f) = f', \Rightarrow \text{linear map.}$$

$$G : C(I, \mathbb{R}) \rightarrow \mathbb{R}, \quad G(f) = \int_a^b f(t) dt,$$

- (i) Let $f : V \rightarrow V'$ be an isomorphism of vector spaces over K . Then $f^{-1} : V' \rightarrow V$ is again an isomorphism of vector spaces over K .
- (ii) Let $f : V \rightarrow V'$ and $g : V' \rightarrow V''$ be K -linear maps. Then $g \circ f : V \rightarrow V''$ is a K -linear map.

Proof. Homework.

↳ "2-line proof".

Let $f : V \rightarrow V'$ be a K -linear map. Then the set

Introduce a subset

$$\text{Ker } f = \{\underline{v} \in V \mid f(v) = 0'\} \quad \text{KERNEL/NUCLEU}$$

is called the **kernel** (or the *null space*) of the K -linear map f and the set

↓
does not make sense for
any function → needs a structure.

$$\text{Im } f = \{f(v) \mid v \in V\}$$

is called the **image** (or the *range space*) of the K -linear map f .

↓
does make sense for any function

Let $f : V \rightarrow V'$ be a K -linear map. Then $\text{Ker } f \leq V$ and $\text{Im } f \leq V'$.
 subspace of V subspace of V'

Proof. We have $f(0) = 0'$, hence $0 \in \text{Ker } f \neq \emptyset$. Let $k_1, k_2 \in K$ and $v_1, v_2 \in \text{Ker } f$. We prove that $k_1 v_1 + k_2 v_2 \in \text{Ker } f$. We have:

$$f(k_1 v_1 + k_2 v_2) = k_1 f(v_1) + k_2 f(v_2) = k_1 \cdot 0' + k_2 \cdot 0' = 0', \quad \checkmark$$

$\nRightarrow (k_1 v_1 + k_2 v_2) \neq 0'$
 $\hookrightarrow f$ of a linear comb. = the linear comb. of the images.

and thus $k_1 v_1 + k_2 v_2 \in \text{Ker } f$. Hence $\text{Ker } f \leq V$.

Now note that $0' = f(0) \in \text{Im } f \neq \emptyset$. Let $k_1, k_2 \in K$ and $v'_1, v'_2 \in \text{Im } f$. We prove that $k_1 v'_1 + k_2 v'_2 \in \text{Im } f$. We have $v'_1 = f(v_1)$ and $v'_2 = f(v_2)$ for some $v_1, v_2 \in V$. Then:

$$k_1 v'_1 + k_2 v'_2 = k_1 f(v_1) + k_2 f(v_2) = f(k_1 v_1 + k_2 v_2) \in \text{Im } f.$$

$\in \text{Im } f?$ $\in \text{Im } f?$

$\Rightarrow \nexists$ linear map. $\Rightarrow$

$\Rightarrow$

$\Rightarrow$

function - surjective (If $\dim V = 0$) surj if kernel is the max. possible.
Kernel of $f \rightarrow$ a linear map is inj. if kernel is the min. possible

Let $f : V \rightarrow V'$ be a K -linear map. Then

$$\text{Ker } f = \{0\} \iff f \text{ is injective.}$$

Proof. $\implies$ Assume that $\text{Ker } f = \{0\}$. Let $v_1, v_2 \in V$ be such that $f(v_1) = f(v_2)$. Then $f(v_1 - v_2) = f(v_1) - f(v_2) = 0'$, hence $v_1 - v_2 \in \text{Ker } f = \{0\}$, and thus $v_1 = v_2$. Therefore, f is injective.

$\impliedby$ Assume that f is injective. Clearly, we have $\{0\} \subseteq \text{Ker } f$. Now let $v \in \text{Ker } f$. Then $f(v) = 0' = f(0)$. By the injectivity of f , we deduce that $v = 0$. Thus $\text{Ker } f \subseteq \{0\}$, and consequently, $\text{Ker } f = \{0\}$. Double inclusion

INJ - SURJ

KERNEL - IMAGE

□

Let $f : V \rightarrow V'$ be a K -linear map and let $X \subseteq V$. Then

How is the linear map behaving with respect to the generating subspaces.

nicely $\leftarrow f(\langle X \rangle) = \langle f(X) \rangle$. applying f to each elem?

Proof. If $X = \emptyset$, then we have:

$$f(\langle \emptyset \rangle) = f(\{0\}) = \{f(0)\} = \{0'\} = \langle \emptyset \rangle = \langle f(\emptyset) \rangle.$$

If $X \neq \emptyset$, use

$\rightarrow$ set of all finite linear combinations

$$\langle X \rangle = \{k_1 v_1 + \dots + k_n v_n \mid k_i \in K, v_i \in X, i = 1, \dots, n, n \in \mathbb{N}^*\}.$$

Since f is a K -linear map, it follows that: apply f to each elem.

$$\begin{aligned} f(\langle X \rangle) &= \{f(k_1 v_1 + \dots + k_n v_n) \mid k_i \in K, v_i \in X, i = 1, \dots, n, n \in \mathbb{N}^*\} \\ &= \{k_1 f(v_1) + \dots + k_n f(v_n) \mid k_i \in K, v_i \in X, i = 1, \dots, n, n \in \mathbb{N}^*\} \\ &= \langle f(X) \rangle, \text{ the set of all linear comb. with those sth's from } X. \end{aligned}$$

Let V and V' be vector spaces over K . Consider on $\text{Hom}_K(V, V')$ the operations: $\forall f, g \in \text{Hom}_K(V, V')$ and $\forall k \in K$, $f + g, k \cdot f \in \text{Hom}_K(V, V')$, where

$\hookrightarrow$ How do I add 2 linear maps?

$$\hookrightarrow (f + g)(v) = f(v) + g(v),$$

$$(kf)(v) = kf(v)$$

$\forall v \in V$. Then $\text{Hom}_K(V, V')$ is a vector space over K .

Let V be a vector space over K . Then $\text{End}_K(V)$ is a vector space over K . the set of all endomorphisms is itself a vector space.

A black-and-white image of (say)

$$n = 1024 \times 768$$

pixels can be viewed as a vector in the real canonical vector space $\mathbb{R}^n$, where each component of the vector is the intensity of the corresponding pixel.

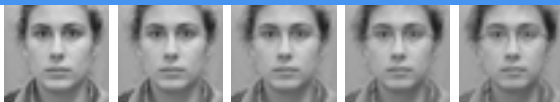
components

Let us consider two vectors representing images:

$$v_1 = \text{img1}, \quad v_2 = \text{img2}.$$

Now consider the following intermediate images:

Start
(m)



Finish (m2)

The vectors corresponding to the above images are the following linear combinations of the vectors v_1 and v_2 :

start

scalar decreasing

$$v_1, \quad \frac{8}{9}v_1 + \frac{1}{9}v_2, \quad \frac{7}{9}v_1 + \frac{2}{9}v_2, \quad \frac{6}{9}v_1 + \frac{3}{9}v_2, \quad \frac{5}{9}v_1 + \frac{4}{9}v_2,$$

increasing

$$\frac{4}{9}v_1 + \frac{5}{9}v_2, \quad \frac{3}{9}v_1 + \frac{6}{9}v_2, \quad \frac{2}{9}v_1 + \frac{7}{9}v_2, \quad \frac{1}{9}v_1 + \frac{8}{9}v_2, \quad v_2.$$

finish

One may use these images as frames in a video in order to get a